Prepared for Bakers & Makers

Executive Summary

This report analyzes sales performance, product demand, zero-revenue transactions, and advertising efficiency to support practical commercial decisions.

Business Questions

- When do weekends generate the most revenue?
- Which products should be prioritized for Saturday sales?
- What patterns may explain zero-revenue transactions used as a proxy for product returns?

Key Highlights

Weekend peak	Saturday Night is the strongest weekend sales period.
Product performance	Cookies are the top-selling product, while Pastry shows the highest estimated return rate.
Ad efficiency	Higher ad spend does not consistently translate into stronger revenue, indicating optimization opportunities.

1. Weekend Revenue Performance

Revenue by Time Slot on Weekends

- Saturday Night generates approximately $30K in revenue, making it the strongest weekend period.
- Sunday Morning follows at approximately $14K in revenue.
- The weakest periods are Saturday Afternoon and Sunday Night.

Actionable Recommendation

- Increase promotional activity during Saturday Night, including targeted late-evening offers.
- Review operating and marketing costs during lower-performing periods such as Sunday Night.

2. Saturday Product Forecast

Forecasted Product Performance

Product	Revenue	Conversions
Cookies	$126,959.82	3,410
Bread	$115,090.16	3,410
Pastry	$126,738.47	3,265

Forecast Caveat

- The forecast shows Cookies declining to near-zero sales in future periods, which conflicts with current performance trends.
- Review seasonality settings and validate the forecast before using it for operational decisions.

Actionable Recommendation

- Prioritize Cookies during high-conversion periods such as Saturday evenings.
- Reassess the forecasting configuration with weekend seasonality and compare projected values against actual results.

3. Zero-Revenue Transaction Review

Methodological Note

The dataset does not contain an explicit return-reason field. Therefore, transactions with Revenue ($) = 0 are used only as a proxy for potential returns. These values should be interpreted as estimated return indicators, not confirmed product returns.

Key Findings

- Pastry records 67 zero-revenue transactions out of 3,265 total conversions, equivalent to an estimated 2.05% rate.
- New customers account for more zero-revenue transactions than returning customers.
- Night periods show frequent zero-revenue events for Bread and Pastry.

Product	Zero-Revenue Count	Total Conversions	Estimated Rate
Pastry	67	3,265	2.05%
Cookies	51	3,410	1.50%
Bread	70	3,410	2.05%

Recommended Actions

- Investigate product freshness and inventory handling, particularly for Pastry.
- Use loyalty incentives to strengthen retention among returning customers.
- Review late-night Bread and Pastry performance before reducing availability; zero-revenue events may indicate low demand or other operational issues.

4. Advertising Efficiency

Ad Spend vs. Zero-Revenue Transactions

- Higher advertising spend does not consistently correspond to higher revenue or fewer zero-revenue transactions.
- One observed Cookies campaign used $47.61 in ad spend and generated $16.56 in revenue, indicating poor short-term return on ad spend.

Actionable Recommendation

- Pause or review campaigns with high spend and low revenue.
- Run A/B tests for underperforming products, especially Pastry, and compare results using consistent performance metrics.

5. Recommendations

Immediate Priorities

1. Collect explicit return data, including a Return Reason field such as Expired, Wrong Item, or Quality Issue.
2. Optimize advertising spend around high-performing products and periods, particularly Cookies and Saturday Night.
3. Investigate freshness, handling, and inventory practices for Pastry.

Long-Term Priorities

1. Track inventory levels to distinguish stockouts from other zero-revenue events.
2. Survey customers to understand why transactions are cancelled or products are returned.
3. Use validated forecasting methods and measure forecast accuracy before relying on projections for planning.

6. Data Limitations

- The dataset does not include explicit return data; zero-revenue transactions are used as a proxy only.
- Advertising spend and zero-revenue transactions may be correlated, but the analysis does not establish causation.

7. Next Steps

1. Refine forecasting with appropriate seasonality and validate results against actual sales.
2. Add slicers for Customer Type, Time of Day, and Date to support interactive analysis.
3. A/B test advertising campaigns for Pastry and Bread and monitor revenue, conversions, and return indicators.